

Judit Campdepadrós Barrios^{a,c}, Sebastian P.H. Speer^{a,b}, Christian Keysers^{a,c}, Cas J.S. Teurlings^a, Ale Smidts^d, Maarten A.S. Boksem^d, Tor D. Wager^e, Valeria Gazzola^a

^a Social Brain Lab, Netherlands Institute for Neuroscience, Amsterdam, The Netherlands

^b Princeton Neuroscience Institute, Princeton University, Princeton, USA

^c Brain and Cognition, Department of Psychology, University of Amsterdam, The Netherlands

^d Rotterdam School of Management, Erasmus University, Rotterdam, The Netherlands

^e Department of Psychological and Brain Sciences, Dartmouth College, Hanover, USA

Introduction

Reward and punishment processing is vital for adaptive behavior, with its dysregulation implicated in mental health and substance use disorders.

Traditional univariate approaches to reward processing often yield small effect sizes and limited reliability. In contrast, **multivariate methods** decode distributed neural patterns, providing higher sensitivity and specificity (Reddan et al. 2017).

In this study (Speer et al. 2023), we developed the **Brain Reward Signature (BRS)**, a multivariate brain model (Kragel et al. 2018) to predict the **magnitude of rewards and losses** in decision-making tasks.

Methods

BRS uses a LASSO-PCR model with data from 5 fMRI tasks across 4 samples (N = 1169).

Procedure:

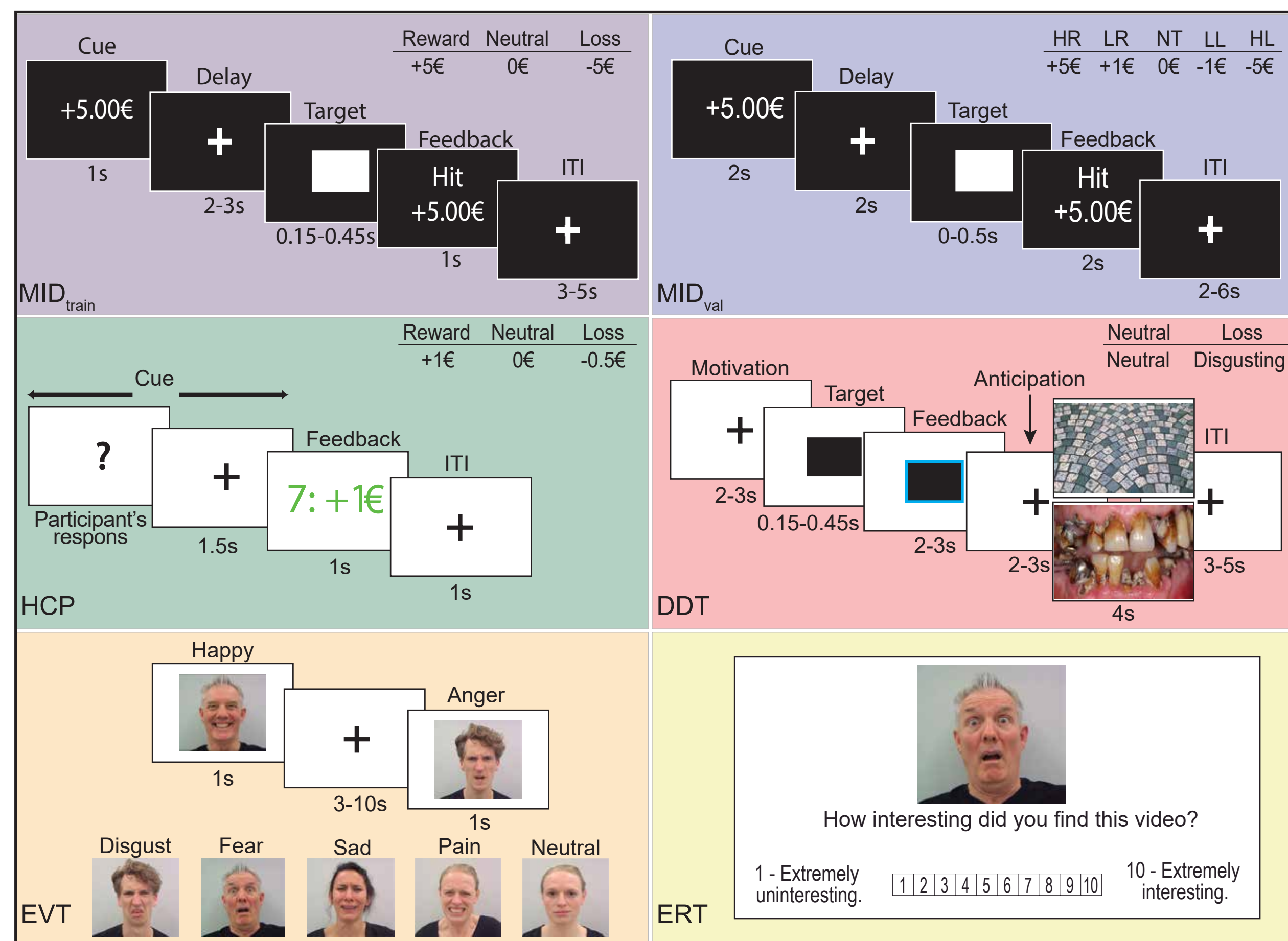
- predict 3 levels of monetary reward
- voxels correlating with outcome, not saliency
- 5-fold cross-validation to prevent overfitting

Training:

- Monetary Incentive Delay task (MID_{train}): N=39

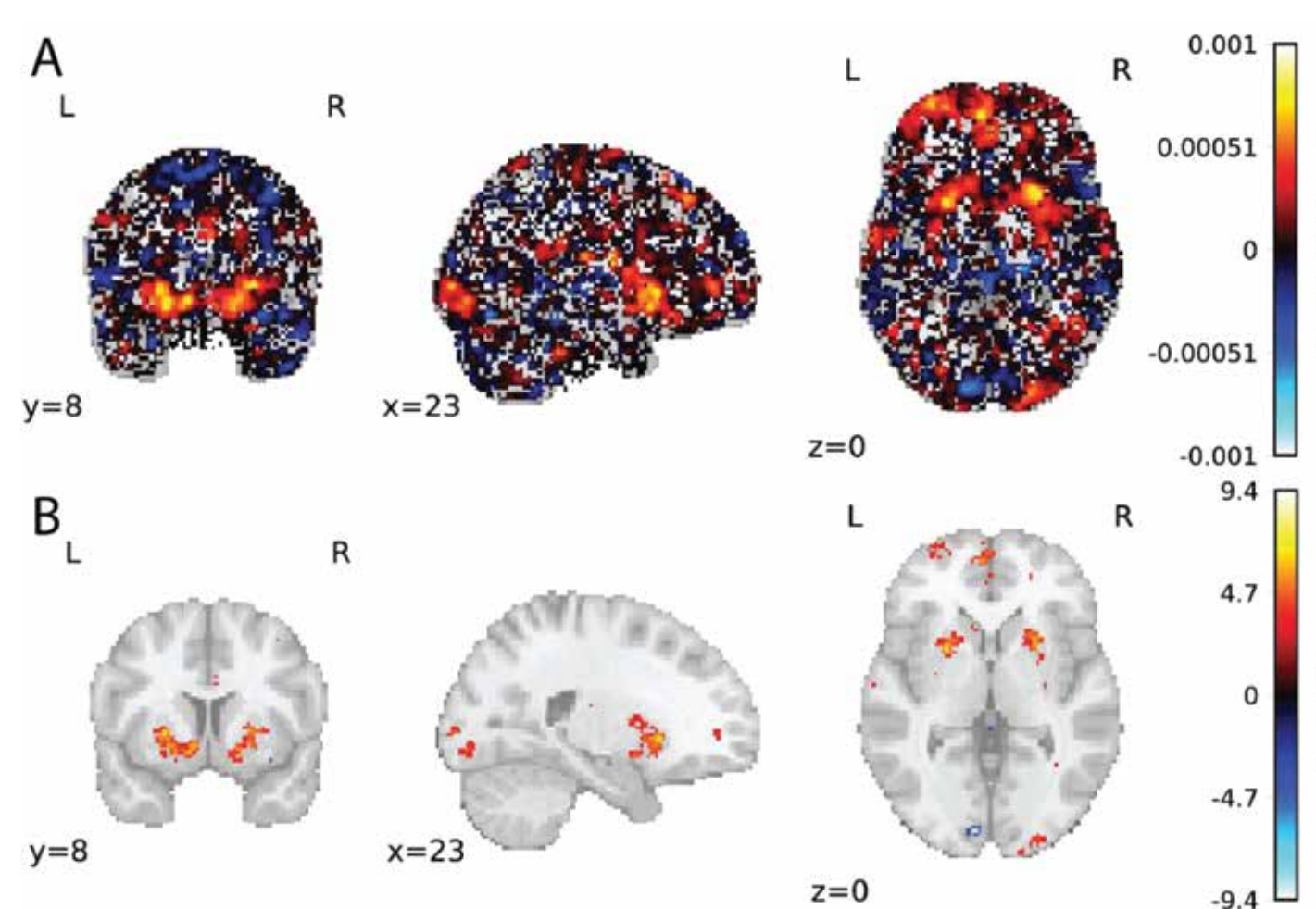
Testing:

- Monetary Incentive Delay task (MID_{val}): N=12
- Gambling task (HCP): N=1084
- Disgust-Delay Task (DDT): N=39
- Emotion Viewing Task (EVT): N=27

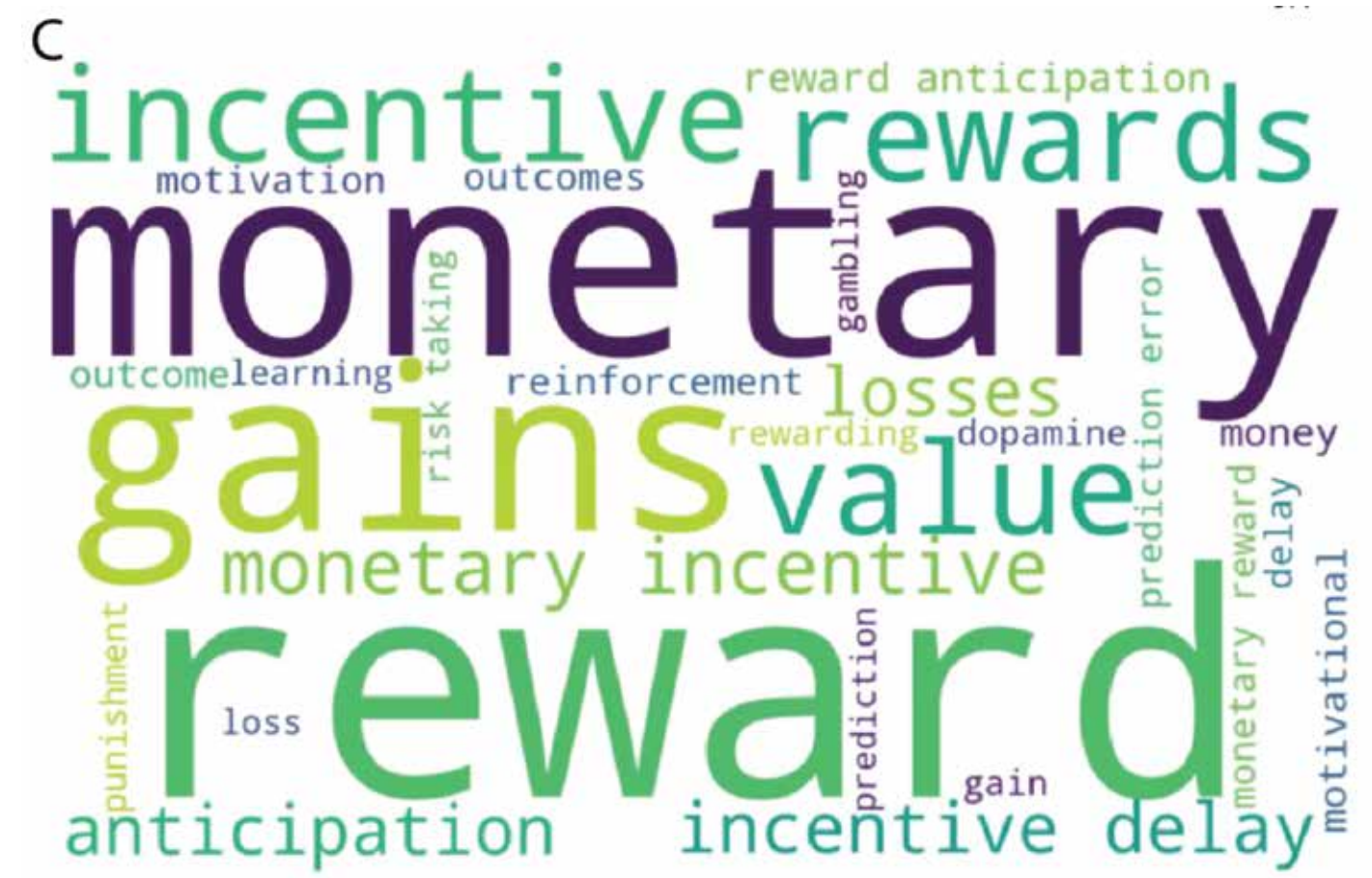


Results

Voxels in dorsal striatum and vmPFC contribute to predictive success of BRS



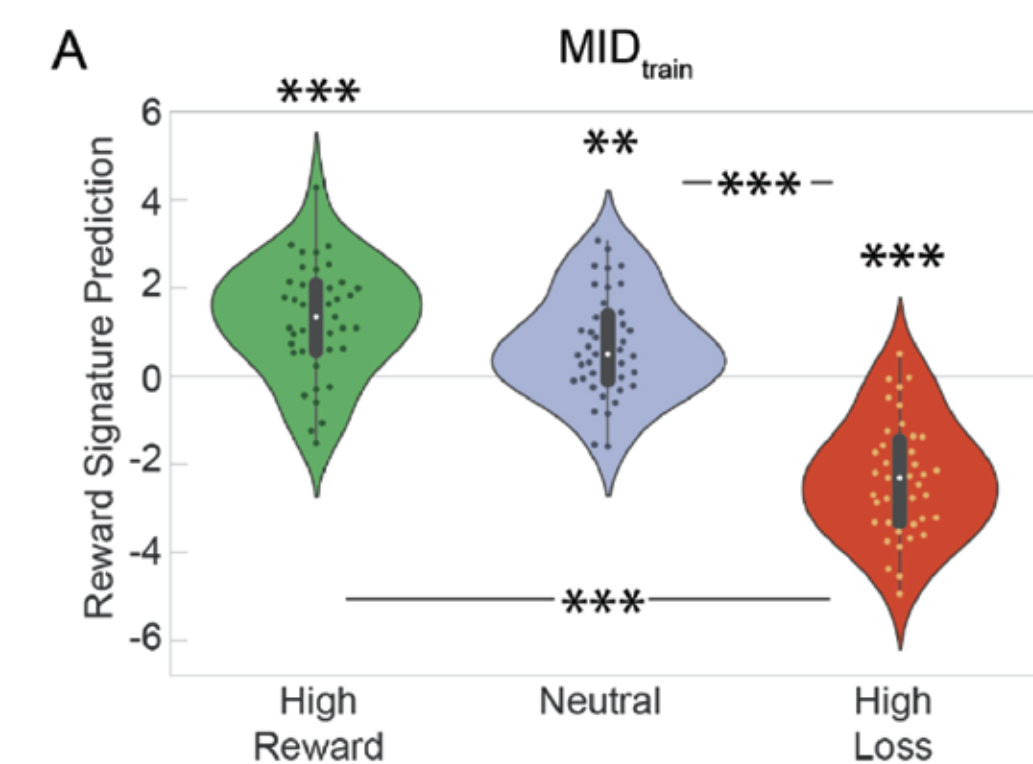
Conceptual word cloud for the meta-analytic decoding of the BRS map



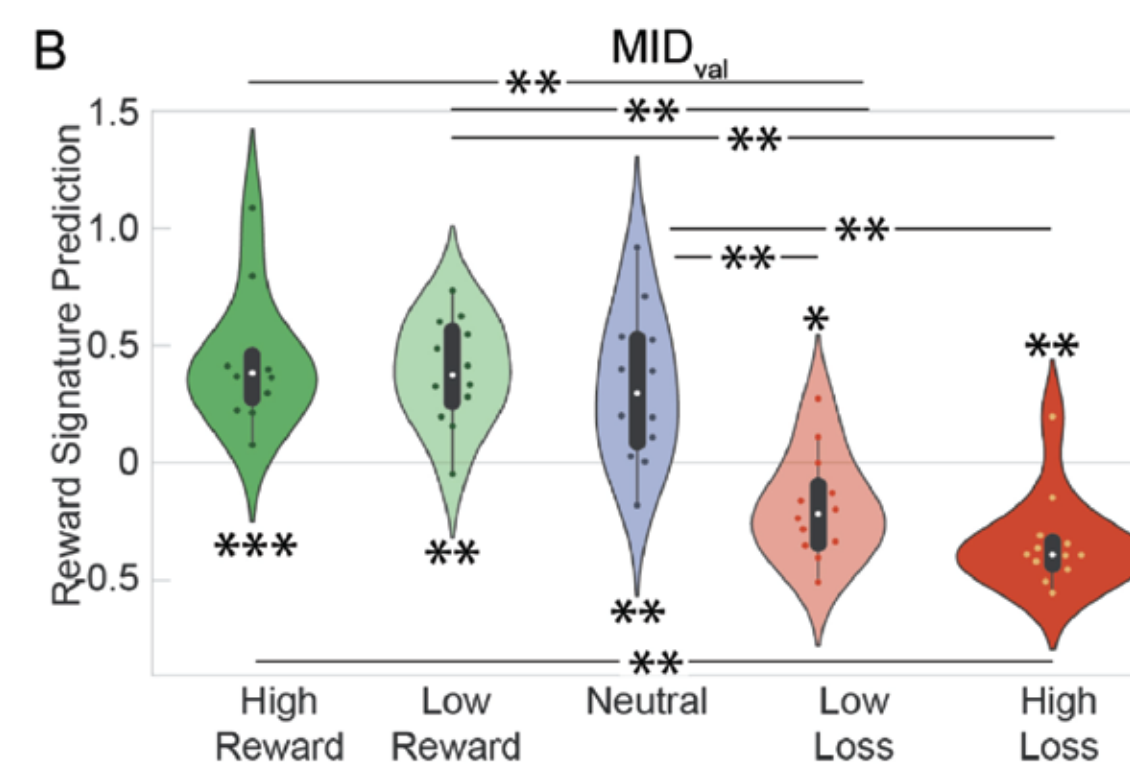
BRS predictions and forced-choice accuracies

Dataset	r	p _{perm}	BF ₁₀	Forced-choice accuracy
MID _{train}	0.72	<.001	> 1000	92%
MID _{val}	0.75	<.001	> 1000	92%
HCP	0.21	<.001	> 100	73%
DID _{outcome}	0.38	<.001	> 100	72%
DID _{feedback}	-0.14	<.280	0.23	--
EVT	all t > 4.5	<.001	> 100	--

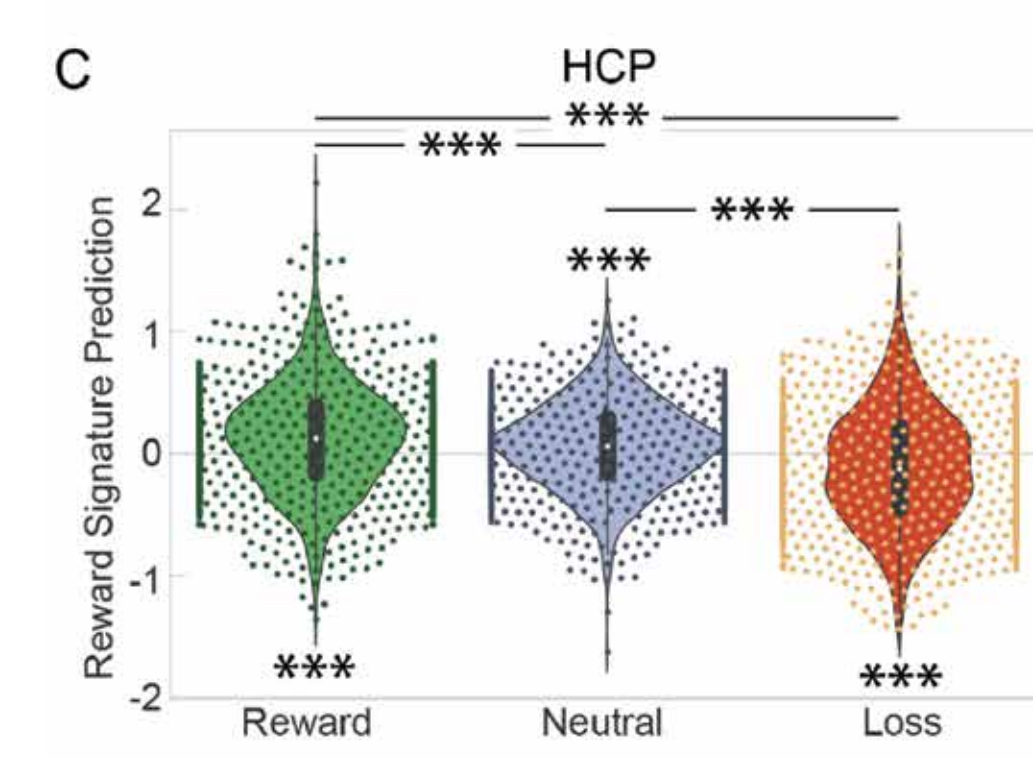
BRS accurately predicts reward in unseen MID participants



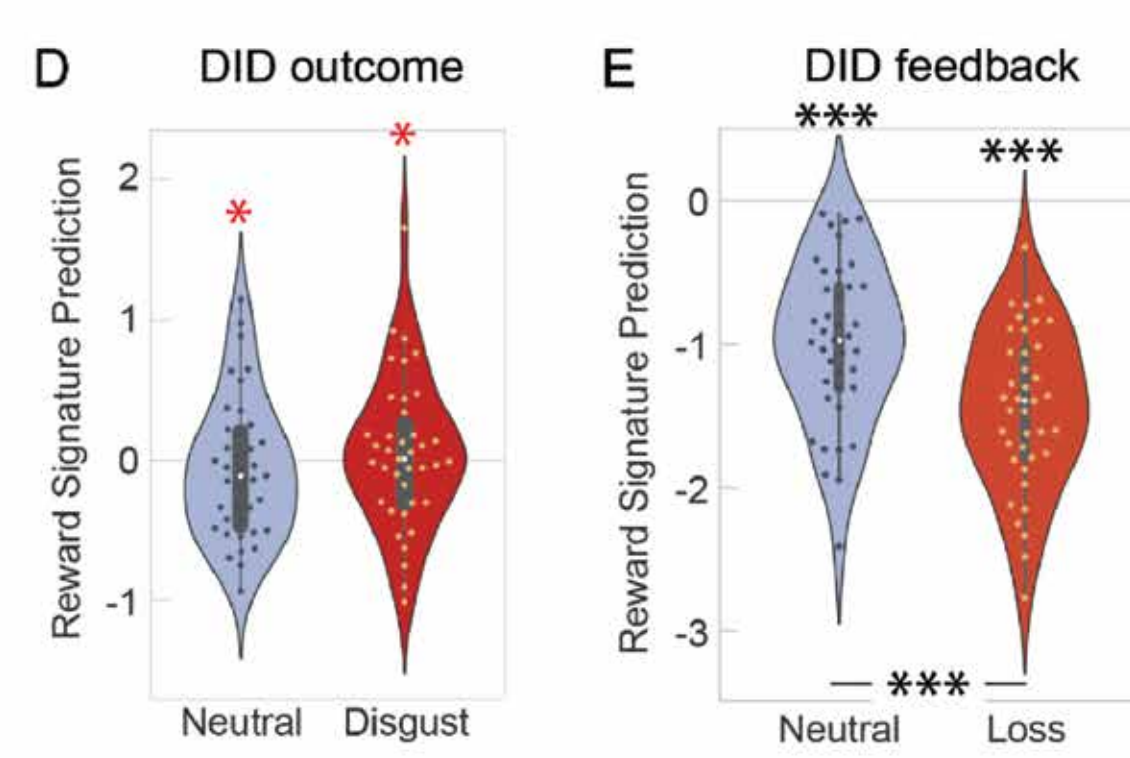
BRS generalizes across task versions and samples



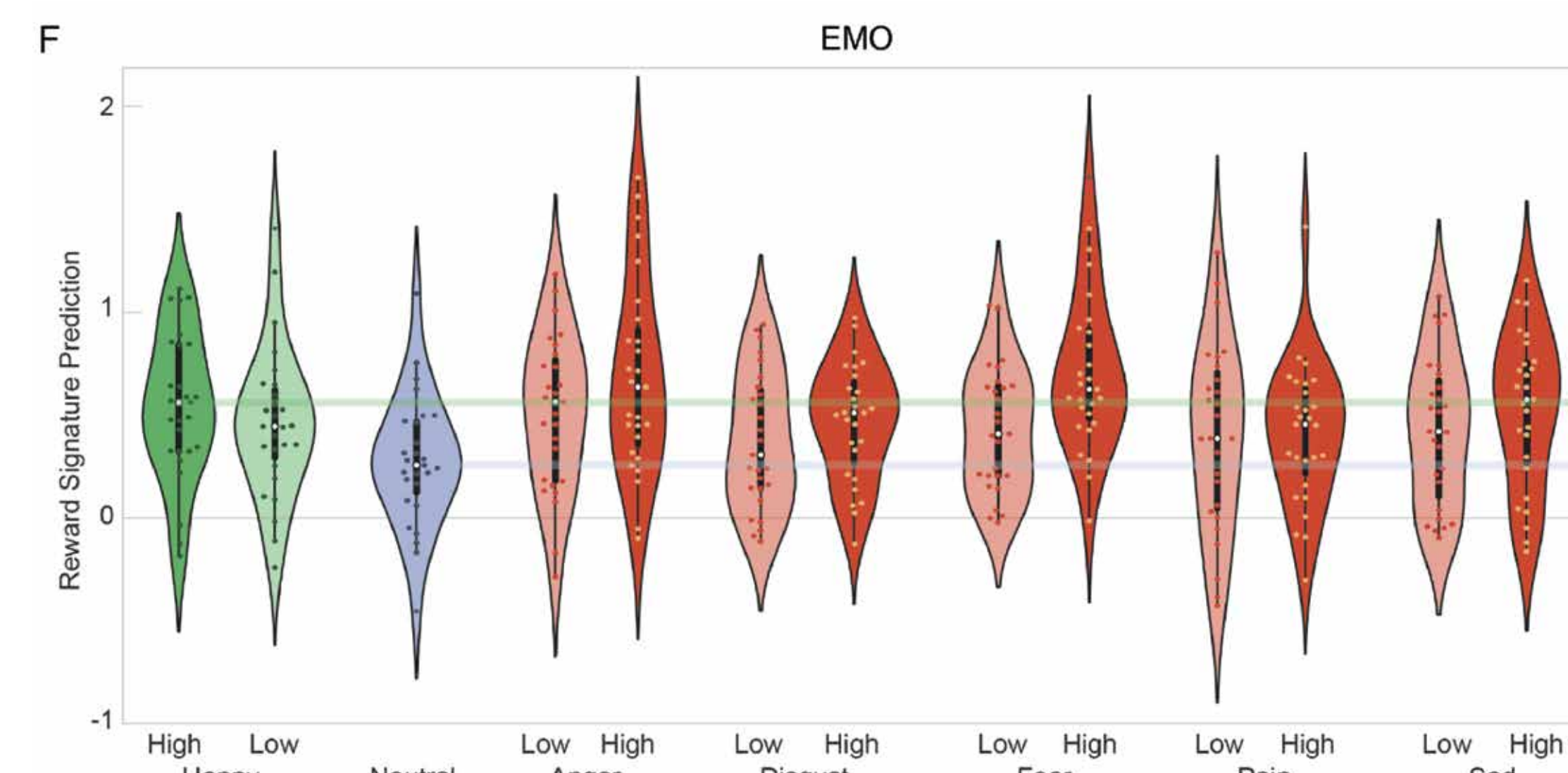
BRS predicts outcomes in a gambling task



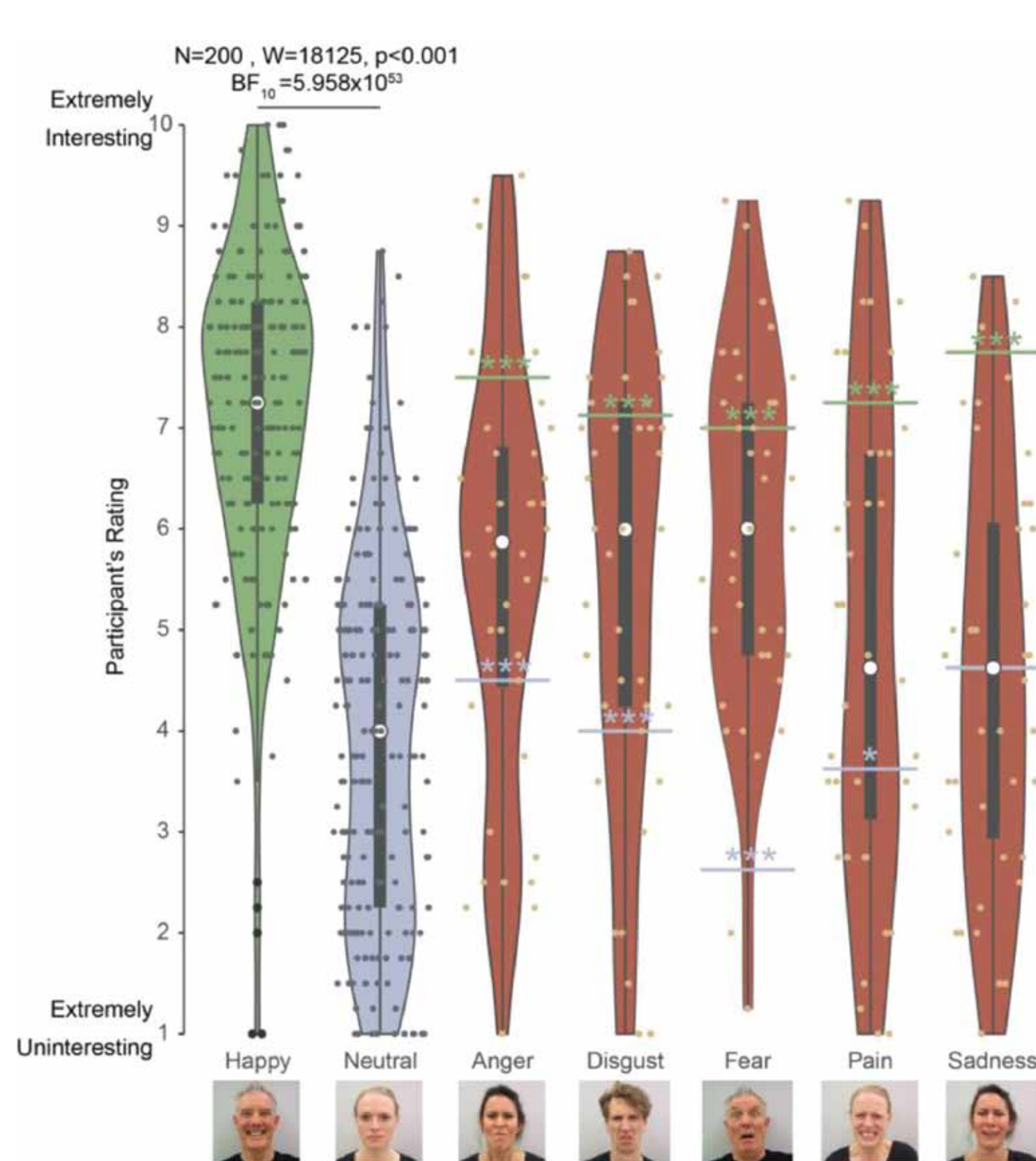
BRS captures non-financial reinforcers, not emotional outcomes



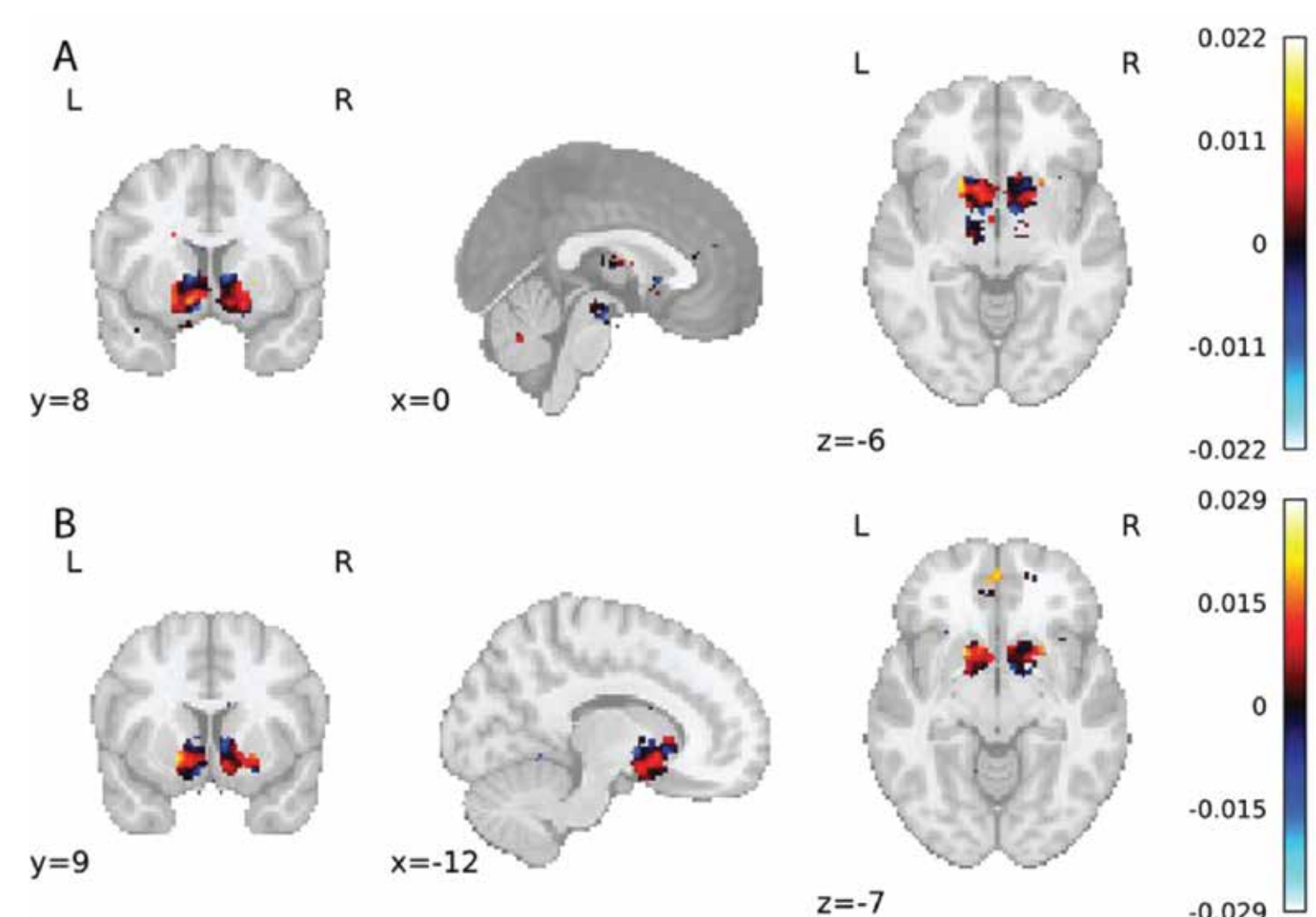
BRS loads on both positive and negative facial expressions, suggesting morbid-curiosity processes (see Oosterwijk, S. et al., 2020)



Positive and negative facial expressions rated as more interesting



Meta-analytic maps on A) monetary reward and B) outcome for feature selection yield similar results



Conclusions

The BRS:

- predicts monetary reward in unseen subjects, samples & tasks,
- generalizes to non-financial reinforcers,
- is specific: does not predict emotional salience, and
- does not generalise when rewards are non action-contingent.

Applications

In a conflictual moral RL task (Fornari et al., 2023), both PE_{shock} and PE_{money} positively loaded on the BRS, showing reward-like responses with less intense than expected shocks and higher than expected monetary rewards.

References

- Fornari, L., Ioumpa, K., Nostro, A. D., Evans, N. J., De Angelis, L., Speer, S. P. H., Paracampo, R., Gallo, S., Spezio, M., Keysers, C., & Gazzola, V. (2023). Neuro-computational mechanisms and individual biases in action-outcome learning under moral conflict. *Nature communications*, 14(1), 1218. <https://doi.org/10.1038/s41467-023-36807-3>
- Kragel, P. A., Koban, L., Barrett, L. F., & Wager, T. D. (2018). Representation, pattern information, and brain signatures: From neurons to neuroimaging. *Neuron*, 99(2), 257–273. <https://doi.org/10.1016/j.neuron.2018.06.009>
- Oosterwijk, S., Snoek, L., Tekoppele, J. et al. (2020). Choosing to view morbid information involves reward circuitry. *Sci Rep* 10, 15291. <https://doi.org/10.1038/s41598-020-71662-y>
- Reddan, M. C., Lindquist, M. A., & Wager, T. D. (2017). Effect size estimation in neuroimaging. *JAMA Psychiatry*, 74(3), 207–208. <https://doi.org/10.1001/jamapsychiatry.2016.3356>
- Speer, S. P. H., Keysers, C., Campdepadrós Barrios, J., Teurlings, C. J. S., Smidts, A., Boksem, M. A. S., Wager, T. D., & Gazzola, V. (2023). A multivariate brain signature for reward. *NeuroImage*, 119990. <https://doi.org/10.1016/j.neuroimage.2023.119990>

Funding

